

PROFILE

Senior Analytics Engineer with an advanced degree in Economics and extensive experience in econometrics, machine learning, and experimentation. Currently completing a Master's in Data Science at the University of Helsinki. Proficient in R, Python, SQL, and GPU computing. Experienced in building production ML models, interactive data applications, and internal analytics tooling.

CONTACT

+358407181510 theo.blauberg@outlook.com [Github](#) [LinkedIn](#)

MASTER'S PROGRAM IN DATA SCIENCE

University of Helsinki

09/2021 -

- **Major:** Data Science
- Specializing in machine learning methods with emphasis on GPU-accelerated computing.
- **Thesis:** Applying tabular foundation models to the detection of fraud patterns in high-dimensional transaction data.

MASTER'S PROGRAM IN ECONOMICS

University of Helsinki

09/2020 - 12/2022

- **Major:** Economics
- **Minors:** Statistics, Mathematics, Computer Science
- **Master's Thesis:** Impact of Policy Shocks in the EU Emissions Trading System on Finland.

EXCHANGE PROGRAM

Fudan University, Shanghai

09/2018 - 06/2019

Studied Chinese language, politics, and economics, with focus on East Asian economic development.

LANGUAGE COURSE

University of Vienna

09/2015 - 12/2015

Intensive German language studies.

Languages: R, Python, SQL, Julia, C++

ML/AI: Torch, tidymodels, scikit-learn, LLMs

Data: dplyr, DuckDB, pandas, Arrow

Web: Shiny, Quarto, PowerBI, REST APIs

- **NVIDIA:** Fundamentals of Deep Learning
- **NVIDIA:** Fundamentals of Accelerated Computing with CUDA Python
- **DataCamp:** Deep Learning in Python
- **DataCamp:** Data Scientist with R

SENIOR ANALYTICS ENGINEER

Nordea

07/2025 -

- Build, maintain, and monitor machine learning models that deliver real-time fraud detection for card and account-to-account transactions.
- Lead analytics initiatives with increased autonomy in model development and deployment decisions.
- Maintain an internal R package for data analysis and visualization, significantly improving team productivity.

DATA ANALYST

Nordea

07/2023 - 07/2025

- Collaborated within a team responsible for building and monitoring real-time fraud detection models.
- Developed internal R packages, Shiny apps, and automated reports. Established a Data & Analytics community within the department.

DATA ANALYST

VATT Institute for Economic Research

01/2023 - 06/2023

- Conducted analyses and co-authored research reports on electricity market data, focusing on [consumer](#) and [company](#) responses to price shocks.
- Designed and developed a [Shiny-based dashboard](#) for interactive data visualization.

RESEARCH ASSISTANT

VATT Institute for Economic Research

08/2022 - 01/2023

- Supported research projects with data preparation, analysis, and visualization tasks.

PROJECT WORKER

City of Tampere

05/2022 - 08/2022

- Analysed a randomised controlled experiment (~5,000 users) testing whether information nudges could shift mobility behaviour, as part of the Keli project (Kestävemmän liikkumisen kehittäminen hiilijalanjälkilaskurin avulla).
- Project co-funded by the Ministry of the Environment. Results published in a [working paper](#).

INTERN

Embassy of Finland in Vienna

05/2021 - 08/2021

- Monitored and reported on Austrian economic developments to inform Finnish Government policy decisions.
- Attended and reported on meetings with UN organizations and local politicians.

CUPLYR - GPU-ACCELERATED DPLYR

- Developing an R package that enables standard dplyr code to execute on GPU hardware through a RAPIDS cuDF backend.
- Implements lazy evaluation with automatic query optimizations, achieving 40-77x speedups over dplyr on large datasets.
- [GitHub](#)

BRACKETEER - TOURNAMENT MANAGEMENT FRAMEWORK

- Developed an R package for modeling and executing tournament competitions with support for multiple formats including round-robin, Swiss system, and elimination brackets.
- Features a pipe-first API design for defining reusable tournament blueprints with automatic stage materialization and flexible result entry.
- [GitHub](#)

DIGITRAFFIC - FINNTRAFFIC API CLIENT

- Building an R package that provides tidy access to Finland's Digitraffic road traffic API, covering 450+ measurement stations with real-time speed, volume, and classification data.
- Features tidyverse-native output, spatial filtering, built-in caching, and rate limiting. To be submitted to CRAN.
- [GitHub](#)

RESEPTOR - AI RECIPE ASSISTANT

- Built a web application for interactive recipe creation using Python Shiny and Claude API integration via Chatlas.
- Features markdown output for easy recipe sharing and distribution.
- [GitHub](#)

PETANQUE LIGA

- Built a [tournament management website](#) with automatic match scheduling and result tracking.
- Backend powered by Google Sheets for multi-device data entry.

IMAGESTOASCII.JL

- Developing a [Julia package](#) for converting images to ASCII art, exploring matrix operations and image processing.

VOTE COUNTER - UNIDO

Authorized by the Western countries group to serve as vote counter in the Secretary-General Election of the United Nations Industrial Development Organization.

BOARD MEMBER - ECONOMICS STUDENTS' ASSOCIATION

Served on the board contributing to strategic planning and student activities.